

Fig. 5.7 The early evolution of the discrete set of initial conditions in their projection onto positions, for $t = 0, 0.2, \ldots, 1.8$, is shown in the *left-most panel*. The subsequent panels show the positions of trajectories at $t = 8, 16, 24$

the nonuniform covering of the area traced out by the pendulum, thus we would not necessarily obtain similar averages until a very long time had passed. Behavior like this is observed in typical molecular models.

5.5 Dynamical Averages

Although the computation of stationary averages is one of the most common uses of molecular dynamics, there is also a need for methods that allow exploration of dynamical processes. We have already seen that the chaotic nature of molecular dynamics means that trajectories are highly sensitive to initial conditions; in the terminology of numerical analysis, such a calculation is *ill-conditioned*. On the other hand it is well known that many molecular systems evolve along sequential *pathways* that can be viewed as the bundles of trajectories emanating from a collection of initial conditions. An example of a dynamical calculation is a transition time, another is the rate of phase space exploration (as measured by the average distance travelled in a given time). These are quite different; the first relies on a spatial localization of the initial and final states; the second is more likely a global average across many states.

In order to define dynamical averages and to describe methods to calculate such quantities, we introduce a straightforward approach based on replacing dynamical averages with stationary averages of an observable that incorporates the action of the flow map. Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be mappings from the phase space to the same Euclidean space, and define the temporal correlation of $\boldsymbol{a}$ and $\boldsymbol{b}$ (on the surface of energy E) by

$$\begin{aligned} C_{\boldsymbol{ab}}(\tau) &\stackrel{\text{def}}{=} k\mathrm{Av}_{H=\mathrm{E}}\left(\boldsymbol{a}(\mathscr{F}_t(z)) \cdot \boldsymbol{b}(z)\right) \\ &= kZ_{\mathrm{E}}^{-1} \int_{\Sigma_{\mathrm{E}}} \boldsymbol{a}(\mathscr{F}_t(z)) \cdot \boldsymbol{b}(z)\delta[H - \mathrm{E}]\mathrm{d}\omega, \end{aligned}$$

where k is a normalization constant that is often incorporated to ensure that $C_{\boldsymbol{ab}}(0) = 1$.

Using the notation developed earlier for the propagator, this has the alternative formulation

$$C_{\boldsymbol{ab}}(t) = k\mathrm{Av}_{H=\mathrm{E}}\left(\boldsymbol{b}(\boldsymbol{z}) \cdot (\mathrm{e}^{t\mathcal{L}}\boldsymbol{a})(\boldsymbol{z})\right).$$

If the system is ergodic, we could write this as a temporal average with respect to any trajectory initiated on the $H = \mathrm{E}$ energy surface:

$$C_{\boldsymbol{a},\boldsymbol{b}}(t) = k \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^\tau \boldsymbol{a}(\boldsymbol{z}(s+t)) \cdot \boldsymbol{b}(\boldsymbol{z}(s))\mathrm{d}s.$$

The *velocity autocorrelation function* is often written in the shorthand notation

$$C_{\boldsymbol{v},\boldsymbol{v}}(t) \stackrel{\text{def}}{=} \frac{\langle \boldsymbol{v}(t) \cdot \boldsymbol{v}(0)\rangle}{\langle \boldsymbol{v}(0) \cdot \boldsymbol{v}(0)\rangle},$$

where the averaging may be assumed to be against the microcanonical measure, if otherwise unspecified.

Now let $\mathcal{A}$ and $\mathcal{B}$ be subsets of configuration space (i.e. subsets in the position space) of nonzero measure, and define

$$T_{\mathcal{A}\mathcal{B}}(\tau) \stackrel{\text{def}}{=} C_{\mathbf{1}_{\mathcal{B}},\mathbf{1}_{\mathcal{A}}}(\tau),$$

where $\chi_{\mathcal{S}}$ is the *characteristic* or *indicator* function of the set $\mathcal{S}$ defined by

$$\mathbf{1}_{\mathcal{S}}(\boldsymbol{z}) = \begin{cases} 1, \boldsymbol{z} \in \mathcal{S} \\ 0, \boldsymbol{z} \in \bar{\mathcal{S}} \end{cases}.$$

We may understand $T_{\mathcal{A}\mathcal{B}}$ as quantifying the number of trajectories which, started from set $\mathcal{A}$ at time $t = 0$, transition to set $\mathcal{B}$ at time τ. Dividing $T_{\mathcal{A}\mathcal{B}}$ by the measure of the set $\mathcal{A}$, we obtain the probability for a trajectory to make this transition in the given time, which we may write as

$$\Pr\left(\mathcal{A} \to_\tau \mathcal{B}\right) = \frac{\langle \mathbf{1}_{\mathcal{B}}(\boldsymbol{z}(\tau))\mathbf{1}_{\mathcal{A}}(\boldsymbol{z}(0))\rangle}{\langle \mathbf{1}_{\mathcal{A}}(\boldsymbol{z}(0))\rangle},$$

where the averaging is assumed to be with respect to the microcanonical distribution at given energy E.